File Edit View Git Project Build Debug Test Tools Extensions Window Help Search LibraryManagementSystem

Sign in

Debug Any CPU LibraryManagementSystem

Test1.cs Library.cs Borrower.cs Book.cs LibraryMa...entSystem

```
<Project Sdk="Microsoft.NET.Sdk">

  <PropertyGroup>
    <TargetFramework>net10.0</TargetFramework>
    <ImplicitUsings>enable</ImplicitUsings>
    <Nullable>enable</Nullable>
  </PropertyGroup>

</Project>
```

100 % No issues found Ln: 10, Ch: 1 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 10:03 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 10:03 AM and took 00.078 seconds ==========
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'LibraryManagementSystem' (2 of 2 projects)

- LibraryManagementSystem
  - Dependencies
  - Book.cs
  - Borrower.cs
  - Library.cs
- LibraryManagementSystem.Tests
  - Dependencies
  - MSTestSettings.cs
  - Test1.cs

Solution Explorer Git Changes

Properties

XML Document

Misc

Encoding	Unicode (UTF-8)
Output	
Schemas	"C:\Program Files\Microsoft Visua
Stylesheet	

Encoding

Character encoding of the document.

Power and sleep settings

Ready Test Explorer Add to Source Control Select Repository

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search LibraryManagementSystem

Sign in

Debug Any CPU LibraryManagementSystem

Test1.cs Library.cs Borrower.cs Book.cs*

LibraryManagementSystem Book Book(string title, string author, string isbn)

```
using System;
11 references
public class Book
{
    1 reference
    public string Title { get; set; }
    1 reference
    public string Author { get; set; }
    3 references
    public string ISBN { get; set; }
    7 references  2/2 passing
    public bool IsBorrowed { get; private set; }
    1 reference
    public Book(string title, string author, string isbn)
    {
        Title = title;
        Author = author;
        ISBN = isbn;
        IsBorrowed = false;
    }
    1 reference
    public void Borrow()
    {
        if (IsBorrowed)
            throw new InvalidOperationException("Book is already borrowed.");

        IsBorrowed = true;
    }
    1 reference
    public void Return()
```

100 % No issues found Ln: 8, Ch: 58 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 10:03 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 10:03 AM and took 00.078 seconds ==========
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

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  - MSTestSettings.cs
  - Test1.cs

Solution Explorer Git Changes

Properties

Ready Test Explorer Add to Source Control Select Repository

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search LibraryManagementSystem

Sign in

Debug Any CPU LibraryManagementSystem

Test1.cs Library.cs Borrower.cs Book.cs*

LibraryManagementSystem Book Book(string title, string author, string isbn)

```
        public string ISBN { get; set; }
        7 references  2/2 passing
        public bool IsBorrowed { get; private set; }
        1 reference
        public Book(string title, string author, string isbn)
        {
            Title = title;
            Author = author;
            ISBN = isbn;
            IsBorrowed = false;
        }
        1 reference
        public void Borrow()
        {
            if (IsBorrowed)
                throw new InvalidOperationException("Book is already borrowed.");

            IsBorrowed = true;
        }
        1 reference
        public void Return()
        {
            if (!IsBorrowed)
                throw new InvalidOperationException("Book was not borrowed.");

            IsBorrowed = false;
        }
    }
```

100 % No issues found Ln: 8, Ch: 58 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 10:03 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 10:03 AM and took 00.078 seconds ==========
```

Solution Explorer

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    - Dependencies
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    - Test1.cs

Solution Explorer Git Changes

Properties

Ready Test Explorer Add to Source Control Select Repository

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search LibraryManagementSystem

Sign in

Debug Any CPU LibraryManagementSystem

Test1.cs Library.cs Borrower.cs* Book.cs*

LibraryManagementSystem Borrower Name

```
using System.Collections.Generic;
7 references
public class Borrower
{
    1 reference
    public string Name { get; set; }
    3 references
    public string LibraryCardNumber { get; set; }
    5 references  2/2 passing
    public List<Book> BorrowedBooks { get; private set; }

    1 reference
    public Borrower(string name, string libraryCardNumber)
    {
        Name = name;
        LibraryCardNumber = libraryCardNumber;
        BorrowedBooks = new List<Book>();
    }

    1 reference
    public void BorrowBook(Book book)
    {
        book.Borrow();
        BorrowedBooks.Add(book);
    }

    1 reference
    public void ReturnBook(Book book)
    {
        book.Return();
```

100 % No issues found Ln: 2, Ch: 1 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 10:03 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 10:03 AM and took 00.078 seconds ==========
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'LibraryManagementSystem' (2 of 2 projects)

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  - Library.cs
- LibraryManagementSystem.Tests
  - Dependencies
  - MSTestSettings.cs
  - Test1.cs

Solution Explorer Git Changes

Properties

Ready Test Explorer Add to Source Control Select Repository

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search LibraryManagementSystem

Sign in

Debug Any CPU LibraryManagementSystem

Test1.cs Library.cs Borrower.cs* Book.cs*

LibraryManagementSystem Borrower Name

```
3 references
public string LibraryCardNumber { get; set; }
5 references  2/2 passing
public List<Book> BorrowedBooks { get; private set; }

1 reference
public Borrower(string name, string libraryCardNumber)
{
    Name = name;
    LibraryCardNumber = libraryCardNumber;
    BorrowedBooks = new List<Book>();
}

1 reference
public void BorrowBook(Book book)
{
    book.Borrow();
    BorrowedBooks.Add(book);
}

1 reference
public void ReturnBook(Book book)
{
    book.Return();
    BorrowedBooks.Remove(book);
}
}
```

100 % No issues found Ln: 2, Ch: 1 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 10:03 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 10:03 AM and took 00.078 seconds ==========
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Solution Explorer

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Solution Explorer Git Changes

Properties

Ready Test Explorer Add to Source Control Select Repository

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search LibraryManagementSystem

Sign in

Debug Any CPU LibraryManagementSystem

Test1.cs Library.cs Borrower.cs* Book.cs* LibraryMa...mentSystem

LibraryManagementSystem Library ViewBorrowers()

```
using System;
using System.Collections.Generic;
using System.Linq;

3 references
public class Library
{
    6 references 1/1 passing
    public List<Book> Books { get; private set; }
    6 references 1/1 passing
    public List<Borrower> Borrowers { get; private set; }

    1 reference
    public Library()
    {
        Books = new List<Book>();
        Borrowers = new List<Borrower>();
    }

    3 references 3/3 passing
    public void AddBook(Book book) => Books.Add(book);

    3 references 3/3 passing
    public void RegisterBorrower(Borrower borrower) => Borrowers.Add(borrower);

    2 references 2/2 passing
    public void BorrowBook(string isbn, string libraryCardNumber)
    {
        var book = Books.FirstOrDefault(b => b.ISBN == isbn);
        var borrower = Borrowers.FirstOrDefault(b => b.LibraryCardNumber == libraryCardNumber);
```

100 % No issues found Ln: 44, Ch: 2 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 10:03 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 10:03 AM and took 00.078 seconds ==========
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

- Solution 'LibraryManagementSystem' (2 of 2 projects)
  - LibraryManagementSystem
    - Dependencies
    - Book.cs
    - Borrower.cs
    - Library.cs
  - LibraryManagementSystem.Tests
    - Dependencies
    - MSTestSettings.cs
    - Test1.cs

Solution Explorer Git Changes

Properties

Library.cs File Properties

Advanced	
Build Action	C# compiler
Copy to Output Directory	Do not copy
Custom Tool	
Custom Tool Namespace	
Misc	
File Name	Library.cs
Full Path	C:\Users\Asus\source\repos\LibraryM

Build Action

How the file relates to the build and deployment processes.

Ready Test Explorer Add to Source Control Select Repository

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search LibraryManagementSystem

Test1.cs | Library.cs | Borrower.cs* | Book.cs*

LibraryManagementSystem > Library > ViewBorrowers()

```
            var book = Books.FirstOrDefault(b => b.ISBN == isbn);
            var borrower = Borrowers.FirstOrDefault(b => b.LibraryCardNumber == libraryCardNumber);

            if (book == null || borrower == null)
                throw new InvalidOperationException("Book or Borrower not found.");

            borrower.BorrowBook(book);
        }

        1 reference  1/1 passing
        public void ReturnBook(string isbn, string libraryCardNumber)
        {
            var book = Books.FirstOrDefault(b => b.ISBN == isbn);
            var borrower = Borrowers.FirstOrDefault(b => b.LibraryCardNumber == libraryCardNumber);

            if (book == null || borrower == null)
                throw new InvalidOperationException("Book or Borrower not found.");

            borrower.ReturnBook(book);
        }

        0 references
        public List<Book> ViewBooks() => Books;
        0 references
        public List<Borrower> ViewBorrowers() => Borrowers;
    }
```

100 % No issues found Ln: 44, Ch: 2 SPC CRLF UTF-8 with BOM

Output

Show output from: Build

```
Build started at 10:03 AM...
========== Build: 0 succeeded, 0 failed, 2 up-to-date, 0 skipped ==========
========== Build completed at 10:03 AM and took 00.078 seconds ==========
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

- Solution 'LibraryManagementSystem' (2 of 2 projects)
  - LibraryManagementSystem
    - Dependencies
    - Book.cs
    - Borrower.cs
    - Library.cs
  - LibraryManagementSystem.Tests
    - Dependencies
    - MSTestSettings.cs
    - Test1.cs

Solution Explorer | Git Changes

Properties

Library.cs File Properties

Advanced	
Build Action	C# compiler
Copy to Output Directory	Do not copy
Custom Tool	
Custom Tool Namespace	
Misc	
File Name	Library.cs
Full Path	C:\Users\Asus\source\repos\LibraryM

Build Action
How the file relates to the build and deployment processes.

Ready | Test Explorer | Add to Source Control | Select Repository

Test Explorer

Search (Ctrl+I)

Test run finished: 4 Tests (4 Passed, 0 Failed, 0 Skipped) run in 135 ms

0 Warnings 0 Errors

Test	Duration	Traits	Error Message
LibraryManagementSystem.Tests (4)	196 ms		

Run | Debug | Profile

Group Summary

LibraryManagementSystem.Tests

Tests in group: 4

Total Duration: 196 ms

Outcomes

4 Passed